

Software Requirements Specification (SRS)

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1. Introduction

1.1 Purpose

This document describes the functional and non-functional requirements for the e-commerce system, providing a basis for its design, construction, testing, and maintenance.

1.2 Scope

The scope of this SRS includes user registration, product listing, searching, purchasing, order tracking, management features for admins, and reporting capabilities.

1.3 Definitions, Acronyms, and Abbreviations

- User: A registered individual using the e-commerce system (buyer or seller)
- Admin: A user with additional privileges for managing products, users, and transactions
- Product: A digital or physical item offered for sale on the e-commerce platform

1.4 References

No external systems, tools, or products are referenced in this SRS.

1.5 Overview

The e-commerce system will enable users to register, login, list and purchase products, track orders, and access reporting features. Admin users can manage product listings, user accounts, and transactions.

2. Overall Description

2.1 Product Perspective

The e-commerce system is a web-based application that facilitates online buying and selling of goods and services.

2.2 Product Functions

- User authentication: registration and login functionality
- Product management: listing, searching, filtering, carting, and ordering
- Order tracking: viewing order history and tracking current orders
- Admin management: managing products, users, and transactions
- Reporting: generating sales and revenue reports

2.3 User Characteristics

Users include both buyers and sellers with varying levels of technical proficiency. The system should be easy to navigate for all users.

2.4 Constraints

- Compatibility with modern web browsers
- Secure handling and storage of user data and transactions

2.5 Assumptions and Dependencies

- Reliable internet connectivity
- Use of standard payment processing systems

3. Functional Requirements

- REQ-001: The system should allow users to register and login using their email and password.
- REQ-002: Sellers should be able to list products with details such as price, description, and images.
- REQ-003: Buyers should be able to search and filter products by category, price, and ratings.
- REQ-004: The system should allow users to add products to a cart and place orders.
- REQ-005: The system should support secure payment processing.
- REQ-006: Users should be able to track their orders and view order history.
- REQ-007: Admin users should be able to manage products, users, and transactions.
- REQ-008: The system should generate sales and revenue reports.
- REQ-009: The system should support at least 800 concurrent users.
- REQ-010: All transactions must be securely handled and recorded.

4. Non-Functional Requirements

4.1 Performance Requirements

No specific performance requirements were provided.

4.2 Security Requirements

The system should enforce strong password policies, use encryption for data transmission and storage, and implement secure payment processing methods.

4.3 Usability Requirements

The system should have an intuitive interface that is easy to navigate for all users.

4.4 Reliability Requirements

The system should be designed to minimize downtime and handle errors gracefully.

4.5 Maintainability Requirements

The system should be modular, allowing for easy updates and expansions.

4.6 Portability Requirements

The system should be web-based, ensuring compatibility with various devices and operating systems.

5. External Interface Requirements

5.1 User Interfaces

- Web browsers: Modern web browsers supporting HTML5, CSS3, and JavaScript

5.2 Hardware Interfaces

No specific hardware interfaces were provided.

5.3 Software Interfaces

Payment processing systems: Integration with standard payment gateways for secure transactions

5.4 Communication Interfaces

Internet connectivity: The system requires reliable internet connectivity to function properly.

6. System Constraints

- Compatibility with modern web browsers and devices
- Secure handling and storage of user data and transactions

7. Assumptions and Dependencies

- Reliable internet connectivity
- Use of standard payment processing systems
- User awareness of the need for a secure connection when accessing the system